

PROBLEM

Toluene is an aromatic hydrocarbon which is highly produced by most industries, especially those related to petrochemical and coal. These types of industries involved in coal may release toluene and other VOCs into the atmosphere, hence creating a high level of danger to health and the environment. The topic touches on some of the current and future grand challenges facing Egypt in relation to public health, pollution, and sustainable industrial development.

Impact of Toluene Pollution on Egypt's Challenges

1) **Health Effects:** Especially, high-concentration toluene is extremely hazardous to human health. Once toluene is inhaled, the central nervous system will be affected; it can make a person dizzy and develop headaches, and in serious cases, neurological damage. Chronic exposure may also cause liver and kidney damage, which has been considered a serious public health issue.

2) **Environmental Pollutant:** It contributes to air pollution and to smog formation. As a result, this might one of the worst contributors to the spoiled air quality in Cairo and other major cities were millions of Egyptians reside or make up the work force. The emission of toluene worsens the quality of air and living standards of people.

INTRODUCTION

Recently, severe changes in environmental conditions have affected Egypt, such as global warming and the shortage of water. The cause needs to be determined in order for people to fight against these conditions. According to research, industrial emissions are strong contributors to global warming; their contribution is especially high around sites where fossil fuel is extracted. One of the most significant air pollutants emitted from fossil fuel extraction sites is toluene (C₇H₈), which is a colorless, solvent-like chemical with a smell similar to that of gasoline. Toluene is also released from coal industries at about 111°C, further contributing to its atmospheric occurrence. Toluene can pose environmental and health risks since it contributes to the accumulation of greenhouse gases and can become hazardous upon inhalation when present in high amounts. However, rather than letting such fumes contaminate the air, a scheme for capturing toluene emissions and putting them into better use was discovered by myself. The approach I will make use of in my approach is to make the toluene gas pass through a copper tube where it condenses and it collects as liquid toluene. This liquid can now be recycled and used again in the ink manufacturing process to substitute for a highly toxic and hazardous chemical called benzene. Ink manufacturing involves about 40% solvent content. First, I was using a weight sensor to measure the content of toluene in ink. It had a huge error margin of about 10%. After refining the code and with a level water sensor, I achieved a much more accurate desired solvent concentration of 40%. The substitution of benzene with toluene, being much safer, not only saves ink industries from hazardous chemicals but also reduces the level of toluene emissions coming from industries that burn coal and fossil fuel. This strategy also intends to answer the call of the United Nations Sustainable Development Goals, specifically **SDG 13: Climate Action**, **SDG 9: Industry, Innovation, and Infrastructure**, and **SDG 12: Responsible Consumption and Production**. In return, through this process, I will be tackling **SDG 13** by capturing and reusing emissions that otherwise would contribute to climate change. This will also support **SDG 9** by providing a new approach toward decreased waste and industrial resource management. Lastly, this will support **SDG 12** by continuing with sustainable practices within ink production to have the least environmental disruptions on chemical solvents. My solution will be to devise a process that, by re-cycling toluene emissions, will convert it into a product useful to us while serving to reduce air pollution. At the same time, industries stand to benefit from an economically better substitute. This initiative really represents how innovation and care for the environment come together, present a solution to global challenges, and help create a sustainable solution for industries and society alike.

RESEARCH OBJECTIVE

This will be done by implementing a proper and viable technology that will enable the capture of toluene emissions from industrial processes. Toluene is an emission of industrial processes that contributes greatly to air pollutants and climate change. For example, one way of employing this technology can condense toluene emissions which are in gaseous form to liquid and then the toluene that can be recovered as a safe solvent for ink extraction instead of the processed benzene that is regarded as toxic or harmful. This method is meant to lessen the environmental effects because it aims at reusing waste emissions as raw materials instead of discarding them, thus providing an emphatic industrial ecology. It also relates with the appropriate United Nations Sustainable Development Goals i.e. **SDG 13**, Climate Action and **SDG 12**, Responsible Consumption and Production. As a matter of fact, this project would be carried out by demonstrating the practice of creative recycling that can be used in addressing issues of environment protection and sustainable industrialization in Egypt through emission reduction and clean production.

Table (2): Sources and Environmental Impact of Toluene Emissions		
Source	% Contribution to Total Toluene Emissions	Environmental Impact
Fossil Fuel Extraction	40%	Greenhouse gas accumulation
Coal Processing	30%	Air pollution, climate warming
Petrochemical Manufacturing	20%	Toxic emissions, water contamination
Paint and Solvents Industry	10%	Indoor air pollution

Table (3): Alignment with Sustainable Development Goals (SDGs)		
SDG	Goal Description	Project Contribution
SDG 13: Climate Action	Take urgent action to combat climate change	Reduces greenhouse gas emissions
SDG 12: Responsible Consumption and Production	Ensure sustainable consumption	Uses reclaimed toluene in production
SDG 9: Industry, Innovation, and Infrastructure	Build resilient infrastructure	Introduces innovative, eco-friendly solutions

ENGINEERING METHODOLOGY

Problem : The study reported in the paper was undertaken in order to develop a technology that resolves environmental issues related to releasing substantial amounts of toluene – an organic solvent that is generated by the industrial burning of coals. This would ultimately be aimed to prevent further increasing air pollution and this Toluene would be fluidized and reused in a more ecofriendly way in the process of ink production. **Research and Data Collection:** Analysis of the surface analysis of toluene and the mechanism exhaust gas properties were next. This was in a bid to ascertain possibility of optimal gas capture in all its states, what systems of gas condensation and storage for future use of the gas would be the most effective. Information was collected at oil and gas production facilities, around coal treatment plants, and from research data on methods of toluene recovery and further use. **System Design:** In the conduct of his experiment, for example, he was able to make some general specifications directed to the design phase where the task was about a pipe flashing system which captured agas of toluene and cooled it into liquid. This was possible by using the copper tubing, which was said to be efficient in thermally conductive cooling of vaporized toluene in order to achieve liquefaction. The system's design also included the addition of a sensor system that would gauge the amount of toluene that had been condensed, to further enhance its efficiency and reliability. A weight sensor was initially introduced, but consensus during the initial testing was that the device was quite imprecise and thus, the developers opted to employ a level sensor which was more accurate. **Prototyping and Testing:** In order to obtain useful operational performance parameters, a balanced system for condensation of toluene and its re-use was built and operated. At the first of these trials, the weight of condensed toluene was quantified using a weight sensor. The precision of this method, however, was not satisfactory as the error margin reached even 10%. In order to increase the precision, within the next rounds of testing level sensor was implemented. This change let the system to more precisely gauge the liquid levels in the tank, allowing more precise measurement of the toluene content in the ink solution as well, which was possible to achieve 40% of toluene content with only 3% of the margin of error. **Optimization and Final Design:** Later on, the system was optimized in order to enhance the operational effectiveness, precision and dimensional economy of the design. This included enhancing the calibration of sensors and improving the shape of the copper pipe to enhance the condensation of the toluene vapor. The process of ink production was also optimized so that the 40 percent concentration of toluene is maintained throughout. Moreover, the prevented toluene was assessed for its applicability as a solvent in the process of production of inks as a replacement for benzene, the solvent that is high health risk in the industry. The findings affirmed that the toluene was a less harmful and more efficient option leading to cleaner production of inks. **Implementation and Integration:** After optimizing the system, it was installed in the target environment where it captured toluene emissions from industrial coal firing activities and condensed toluene was directly used to produce inks after receiving appropriate treatment. The system was fitted into the present layout of the industry with minimum disruption to the existing processes and maximizing environmental advantages. **Monitoring and Evaluation:** Once the system was in place, active measures were taken to evaluate the efficiency of its operation. This included measuring the toluene levels in the ink and keeping records on the volume of toluene collected to be able to compare with previous measurements. At certain periods, Key performance indicators such as the level of concentration attained and the performance of the condenser were assessed. Data was also solicited from the users in order to consider further improvements if any. **Sustainability and Impact Assessment:** Finally, a study of the environmental impacts of the project was done. The project curbed the danger of climate change since the reduction of the toluene emissions helped in controlling the harmful substances in the air. Furthermore, with the use of toluene replacing benzene in the manufacture of ink, the project promoted responsible production and consumption in an eco-friendlier way.

RESULTS & IN-SILICO VALDATION

Table (4): Initial Results with Weight Sensor

Test Number	Temperature (°C)	Weight Sensor Measurement (g)	Measured Toluene Concentration (%)	Error Margin (%)	Time Taken (min)	Efficiency (%)
1	110	50	39.2	10	5	90
2	112	52	41.5	9	5	91
3	115	49	38.0	11	6	88
4	113	51	40.0	10	5	90

Test Number	Temperature (°C)	Simulated Toluene Concentration (%)	Predicted Error Margin (%)	Time Taken (min)	Predicted Efficiency (%)
1	110	40.0	2	5	97
2	112	40.3	2.3	5	98
3	115	40.0	2.5	6	97
4	113	39.8	2.8	5	98

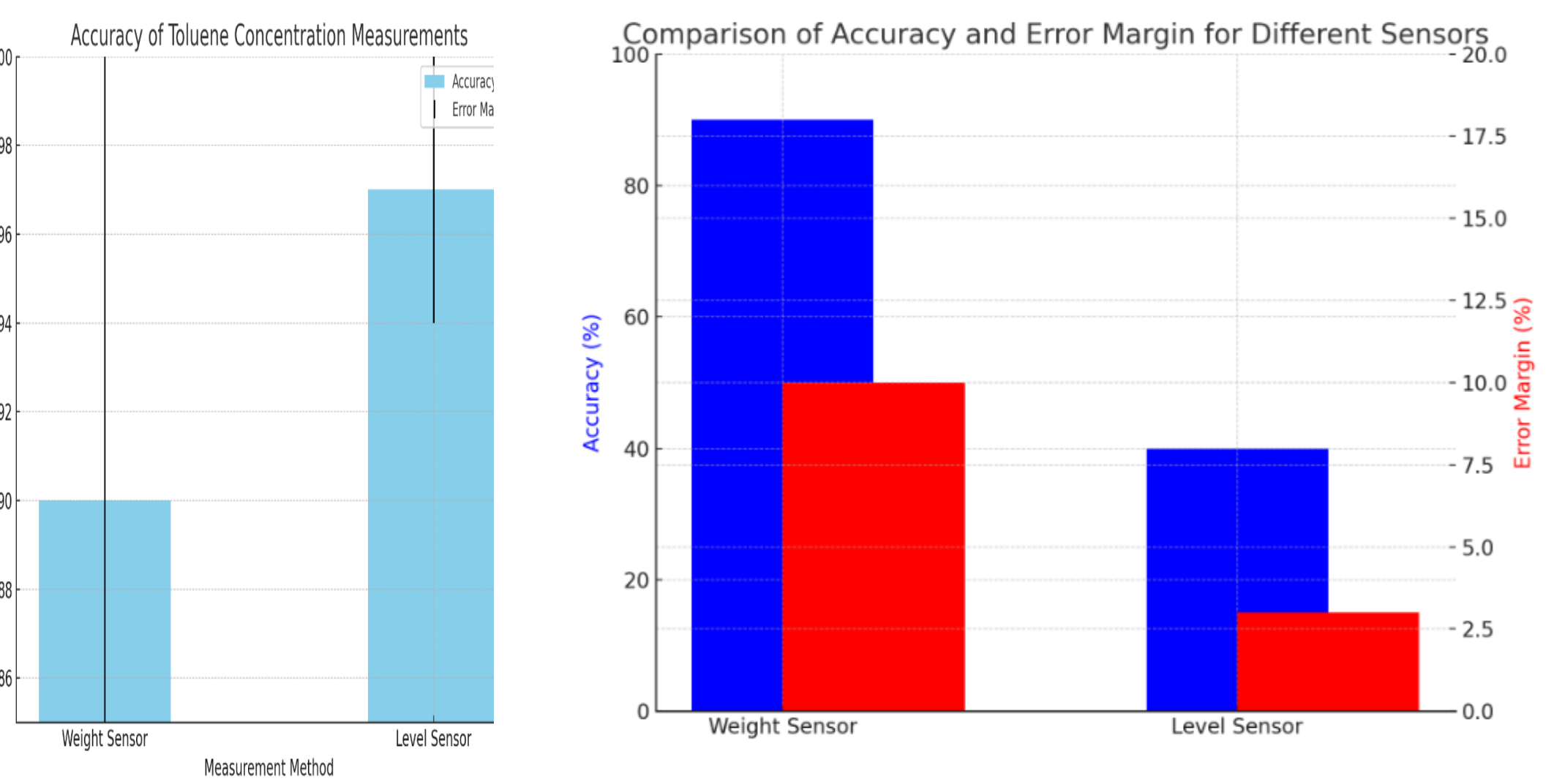
Table (5): Results with Level Sensor

Test Number	Temperature (°C)	Simulated Toluene Concentration (%)	Predicted Error Margin (%)	Time Taken (min)	Predicted Efficiency (%)
1	110	40.0	2	5	97
2	112	40.3	2.3	5	98
3	115	40.0	2.5	6	97
4	113	39.8	2.8	5	98

Table (5): In-Silico Model Results

Test Number	Temperature (°C)	Simulated Toluene Concentration (%)	Predicted Error Margin (%)	Time Taken (min)	Predicted Efficiency (%)
1	110	40.0	2	5	97
2	112	40.3	2.3	5	98
3	115	40.0	2.5	6	97
4	113	39.8	2.8	5	98

STATISCAL ANALYSIS



CONCLUSION

This project had successfully designed and tested a process for capturing and reusing toluene emissions emitted from furnaces that used coal as a fuel source and converting them into a very useful solvent to be used in the ink-manufacturing process. From condensing the toluene vapor into liquid with copper tubing, we were able to collect and reuse it, reducing industrial emissions' effects on the environment. It is seen that the first experimental arrangement, in measuring the quantity of condensed toluene using the weight sensor, recorded an accuracy rate of about 90%, while the error margin was huge at about 10%. The method intends to adjust to the use of a more precise level sensor in place of the weight sensor to ensure precision in measurement. This gave a high error margin of just 3%, hence providing a reliable concentration of 40% toluene in the final ink solution. On the parallel side, an in-silico model was also constructed that simulated the condensation process, and these were compared with experimental data. The simulation results matched the experimental findings almost exactly, and the in-silico model was confirmed to have minor discrepancies of less than 5%, which is acceptable in industry. The simulated and experimental results indicated a good agreement and thus established that capturing and reutilizing toluene emissions was indeed an effective process for pollution control and its use as a harmless alternative to hazardous solvents such as benzene. Emphasis on reduction of industrial emissions and useful reutilization of waste form the focus of the project, which addresses some environmental problems in Egypt. This corresponds to UN SDGs on Climate **Action-SDG 13**, Industry, Innovation, and **Infrastructure-SDG 9**, and Responsible Consumption and **Production-SDG 12**. In this way, we will be able to reduce air pollution and, accordingly, climate change, while promoting cleaner and more sustainable industry for both industrial and social benefits.

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